**Sampling, Frequencies and Signal Reconstruction: From Theory to Practice**

Sampling is one of the fundamental concepts in signal processing. It is the procedure by which an analog continuous-time signal is converted into a discrete-time signal suitable for digital processing. Theoretically, sampling is expressed by the relation:

$x[n] = xa(nTs)$

where $x[n]$ is the sequence of samples and Ts is the sampling period. The corresponding sampling frequency is

$fs = 1/Ts$.

**Analog- and discrete-time frequency**

In analog time, frequency is measured in Hz. In discrete time, however, we do not talk about Hz but about normalized frequency (cycles per sample) or angular frequency per sample:

$fd = f / fs$ ,   $\omega = 2\pi ( f / fs )$

This relation shows that the frequency of the discrete-time signal depends directly on the sampling frequency. However, the physical, analog phenomenon does not change — what changes is how we "see" it through the samples.

**The Nyquist–Shannon theorem**

For a signal to be reconstructed without loss from its samples, the sampling frequency must satisfy

$fs \geq 2fmax$

where fmax is the maximum frequency present in the spectrum of the signal. If this condition is not satisfied, the phenomenon of frequency folding (aliasing) appears, where the signal seems to have another lower frequency than the real one.

In this case, different analog sinusoids produce the same sequence of samples; that is, there exist infinitely many continuous-time signals compatible with the same samples. Only when the Nyquist criterion is satisfied and the signal is bandlimited is the reconstruction unique.

**Aliasing and physical interpretation**

When $fs < 2fmax$:

- too few samples per period are taken
- different analog frequencies coincide in the discrete domain
- the observer sees another sinusoid (alias)

Thus, the signal cannot be recovered correctly, regardless of the interpolation method.

**Signal reconstruction and interpolation**

After sampling, the problem of reconstruction arises. In ideal theory:

[ideal sinc interpolation expression]

This formula describes ideal sinc interpolation and constitutes the only method of perfect reconstruction of ideally bandlimited signals.

In practice, simpler approximations are often used:

- connecting samples by straight-line segments (linear interpolation)
- zero-order hold (keeping the value constant until the next sample)

As fs increases, that is, as we take more samples per period, linear interpolation can achieve any desired accuracy, at least within practically acceptable error. For a given number of decimal digits of accuracy, there always exists a sufficiently high sampling frequency that makes linear interpolation satisfactory.

**Practical systems: DAC, ZOH and filters**

In hardware, after the digital-to-analog converter:

- Zero-Order Hold is applied — the signal becomes “stair-shaped”
- an analog reconstruction low-pass filter follows
  – smooths the steps
  – removes unwanted high harmonics
  – approximates ideal sinc-type reconstruction

Thus, hardware achieves correct reconstruction with lower fs than would be required if we were to perform simple linear interpolation without filtering.

**The spectrum and the role of fmax**

For a general signal, its spectrum is obtained via the Fourier transform. Theoretically, most signals have infinite bandwidth, meaning that the “true” fmax is infinite. However, real systems:

- introduce an analog anti-aliasing filter
- cut frequencies above a cutoff frequency fc
- practically define fmax = fc

Thus, they make the signal artificially bandlimited. This is a conscious approximation: a small, very high-frequency part of the spectrum is sacrificed, which is usually neither perceptible nor useful.

**Error in practice**

The filter is not ideal and therefore the spectral truncation is not perfect. However, the error:

- is controlled
- lies below perceptual thresholds
- is often masked by system noise

In audio applications, for example, the ear does not perceive frequencies above 20 kHz. In telephony, the spectrum is limited even further, without essential loss of speech intelligibility.

**Essential summary**

- Theoretical signals have infinite spectra.
- Practical engineering:
    - keeps the useful part of the spectrum
    - limits it using an anti-aliasing filter
    - selects fs ≥ 2fc
    - reconstructs with a low-pass filter

Thus, high accuracy is achieved without the need for infinite bandwidth or infinite sampling frequency. The small errors are intentional and designed to be imperceptible and practically negligible.

## Selection of sampling frequency in practice and the role of filters

The selection of the sampling frequency fs is a critical stage in the design of digital signal processing systems. Although the theoretical starting point is the Nyquist–Shannon criterion, in practice other factors are also involved, such as filter implementability, energy cost, and the desired reconstruction accuracy.

First, the frequency region of interest is determined, that is, the useful spectrum of the signal. This is the portion of the spectrum where the information we want to preserve is located. Since physical signals are rarely ideally bandlimited, we assume that very high frequencies are either insignificant or practically imperceptible.

Next, before the A/D converter, an analog low-pass anti-aliasing filter with cutoff frequency fc is placed. This filter suppresses high frequencies above fc so that they do not fold into the useful spectrum after sampling. Because practical filters do not have an abrupt cutoff, fc is placed slightly above the region of interest and a transition band is provided.

The sampling frequency is then selected so that

fs ≥ 2fc

with a small additional safety margin (guard band), to take into account the non-ideal shape of the filter. In this way, aliasing is avoided; that is, the overlapping folding of spectral components that would irreversibly distort the signal is prevented.

At the output of a system, after the D/A converter, the signal is not reconstructed by joining samples with straight lines. First, a zero-order hold operation takes place, during which each sample is held constant until the next one, creating a “staircase” waveform. Then an analog low-pass reconstruction filter is applied, which smooths the steps and removes unwanted high frequencies. The action of this filter approximates ideal sinc reconstruction predicted by theory.

An important practical conclusion is that the desired accuracy does not necessarily require an extremely high sampling frequency. Theoretically, if we used a very high fs and simple linear interpolation (straight-line joining), we could achieve an excellent signal approximation. However, this solution entails excessive data volume, increased computational load, and higher energy consumption.

Instead, real systems adopt a more balanced approach: they keep fs at reasonable values and rely on the cooperation of the anti-aliasing filter and the reconstruction filter to achieve high reconstruction quality. Essentially, there is a trade-off between sampling frequency and filter complexity. Better filters allow lower fs, while higher fs allows simpler filters.

Ultimately, the goal is not theoretical perfection but practical accuracy. The errors introduced by spectral truncation or non-ideal reconstruction are chosen to be smaller than the limits of human perception or the system noise. Thus, a system is achieved that operates efficiently, without excessive sampling frequency and computational resources, while maintaining signal quality at the required levels.

## Proper selection of sampling and quantization parameters in digital signal processing systems

In a digital signal processing system, the final object of processing is the discrete sequence x[n], which has resulted from an analog signal after sampling and quantization. The quality and usefulness of x[n] depend decisively on the proper design of the stages preceding processing: the selection of the anti-aliasing filter cutoff frequency fc, the sampling frequency fs, and the quantization characteristics.

### Proper selection of cutoff frequency fc

Before sampling, the analog signal passes through an analog low-pass anti-aliasing filter. The purpose of this filter is to limit the spectrum of the signal to a frequency region that can be represented without aliasing. Since real signals usually have spectra with non-zero tails at high frequencies, the filter essentially cuts off these high components.

The selection of fc is made so that all the information necessary for the application is preserved, while frequencies above fc are considered useless or imperceptible. In this way, the loss of information caused by the filter is controlled and concerns a part of the spectrum that does not need to be represented.

### Selection of sampling frequency fs

After fc is chosen, the sampling frequency is determined according to the Nyquist criterion:

$fs \geq 2fc$

In practice, a slightly larger value is usually selected so that a guard band exists due to the non-ideal slope of the filter. Proper selection of fs ensures that the spectral replicas created by sampling do not overlap and therefore aliasing does not appear in the useful spectrum.

If fs is excessively large, the data volume and computational load increase without substantial quality gain. Conversely, if it is too small, aliasing occurs and useful information is irreversibly lost. Therefore, the optimal choice of fs is the result of a balance between quality and efficiency.

### Quantization and number of bits

Time sampling is accompanied by amplitude quantization. Each sample is not stored with infinite precision but is approximated by the nearest discrete level. The density of these levels is determined by the number of bits of the A/D converter.

Quantization error is unavoidable and manifests itself as additional noise in the signal. However, the number of bits is chosen so that this error is much smaller than the useful signal and often below the noise level of the system itself or human perception thresholds. In this way, the loss of amplitude accuracy does not substantially affect the information.

**Final situation during digital processing**

After the above stages, the system has only the discrete signal x[n]. This contains:

- error due to cutting frequencies above fc
- limitations due to finite fs
- quantization error due to finite bits

However, all these errors have been taken into account during design and concern information that is not useful for the application. Consequently, once x[n] has been created with proper parameters, these losses do not affect the desired information and do not need to concern the digital processing stage.

**Conclusion**

The correct combination of selecting fc, fs and quantization accuracy forms the foundation for reliable digital signal processing. The errors that are inevitably introduced in the preceding stages are designed so as to concern useless or imperceptible information. Thus, in the actual processing phase we work exclusively with x[n], knowing that it contains all the information we need.